

Image formats

Image file $\Rightarrow$ Load $\Rightarrow$ Convert to array $\Rightarrow$ Process $\Rightarrow$ Save

Raster: Image pixellat jadvali.

Vector: Image pixel emas, formula bilan saqlanadi.

ML da Raster dan foydalaniladi.

Raster image formats

Format	Signifik	Alpha	IXlatilish
JPEG	- Lossy	- yuq	- bituratlari, ko'p soxtaxt
PNG	- lossless	- bor	- logotip, screenshot
GIF	- lossless	- oddiy	- pitpe animatsiya
WebP	- Ikkala	- bor	- veb saytlar
HEIC	- lossy	- bor	- iPhone rasmlari
BMP	- Signifika	- odatda yuq	- eski, katta fayl
TIFF	- odatda birlisi	- bor	- tibbiyot, suniy yo'ldosh
RAW	- Signifika	- yuq	- kamerasiz foto

Compression $\Rightarrow$ file hajmini kichraytirish.

Lossless (izotiklik)

Lossy (yotiklik)

- Ochiqanda rasi o'z ushadek qaytadi. Pixel qiymatlari o'zgarib qaytadi.

PNG, GIF, TIFF, BMP.

- Ba'zi ma'lumotlar tarkib quboniladi. Insa u'zi

sozmo'ydiyan mayda farqlar

JPEG, HEIC, WebP

Image to tensors

- ⇒ Image bi Deep Learning input
- ⇒ Input sifatida number kiritilgan kerak
- ⇒ Demak, Image'ni → tensorflow shartlari

1. Image file
↓
PIL Image
↓
Tensor
↓
NN

2. Image file
↓
PIL Image
↓
Numpy Array
↓
Tensor
↓
NN

Number ⇒ 0D tensor

List ⇒ 1D tensor

Table ⇒ 2D tensor

Image ⇒ 3D tensor

(H, W, C) ^{h x w x c} ⇒ Numpy, PIL, OpenCV

(C, H, W) ^{c x h x w} ⇒ Pytorch

Normalization

Pixel qiymatlarini (0 dan 255 gacha)
model uchun qulay oraligga keltirish.

Linear Scaling $[0, 1]$ ga keltirish.

$$x' = \frac{x}{255}$$

$0 \rightarrow 0$, $128 \rightarrow 0.502$, $255 \rightarrow 1$

2-usul: Standardization:

=> Mean orteche biquat,

=> Standard deviation (σ): - biquatlar
ortechedan orteche gancha ushga
ekani kirsatadi.

$$x' = \frac{x - M}{\sigma}$$

=> Hamma bir x' o'choqga keltirish
model terrog va bog'lanishni oshiradi.

=> Overfittingni ham oldi olindi.

Result: (0, 255) $\rightarrow$ (0, 1) normalization

Pytorch (C, W, H) dan foydalanadi,
Asariy sababi samaradorlik va
tezlik uchun.

Feature	Pillow	OpenCV
Open image	Image.open()	cv2.imread()
Format info	.format (JPG, PNG)	Not directly available.
Color mode	.mode (RGB...)	Determined by shape.
Color order	RGB	BGR
Data structure	PIL Image	Numpy array